Lab1 Report Microprocessor

Umesh Motiwale

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1 Problem 1

1.1 Objective

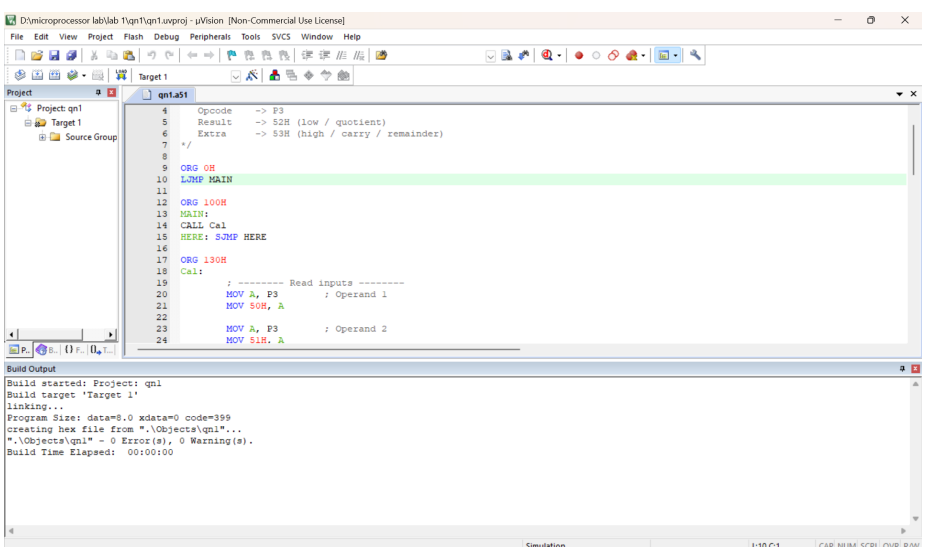
The objective of this experiment is to analyze the given assembly program (Error_code.asm) containing errors, identify and debug all syntax and logical mistakes using Keil µVision and successfully build the corrected program.

1.2 Simulation

1.2.1 Errors

```
Build started: Project: qn1
Build target 'Target 1'
assembling qn1.a51...
qn1.a51(47): error A9: SYNTAX ERROR
qn1.a51(57): error A38: NUMBER OF OPERANDS DOES NOT MATCH INSTRUCTION
qn1.a51(65): error A38: NUMBER OF OPERANDS DOES NOT MATCH INSTRUCTION
qn1.a51(79): error A9: SYNTAX ERROR
Target not created.
Build Time Elapsed:  00:00:00
```

1.2.2 Errors Solved and Build Successfully



```
Opcode    -> P3
Result    -> 52H (low / quotient)
Extra     -> 53H (high / carry / remainder)
*/

ORG 0H
LJMP MAIN

ORG 100H
MAIN:
CALL Cal
HERE: SJMP HERE

ORG 130H
Cal:
        ; -------- Read inputs --------
        MOV A, P3        ; Operand 1
        MOV 50H, A

        MOV A, P3        ; Operand 2
        MOV 51H, A
```

```
Build started: Project: qn1
Build target 'Target 1'
linking...
Program Size: data=8.0 xdata=0 code=399
creating hex file from ".\Objects\qn1"...
".\Objects\qn1" - 0 Error(s), 0 Warning(s).
Build Time Elapsed:  00:00:00
```

1.3 Observation

- The provided program Error_code.asm could not be built initially due to multiple syntax and logical errors.
- Errors like CJNZ was changed to CJNE (Compare and Jump if Not Equal), DIV A,B was changed to DIV AB, MUL A,B was changed to MUL AB, SUB was changed to SUBB.
- After fixing all identified errors, the program compiled successfully without any warnings or errors.
- The corrected program executed as expected in debug mode, confirming proper functionality.

1.4 Conclusion

- The experiment helped in understanding the importance of correct syntax and instruction sequencing in 8051 assembly programming.
- This exercise improved familiarity with Keil µVision debugging tools and reinforced concepts related to assembly language programming and error handling.

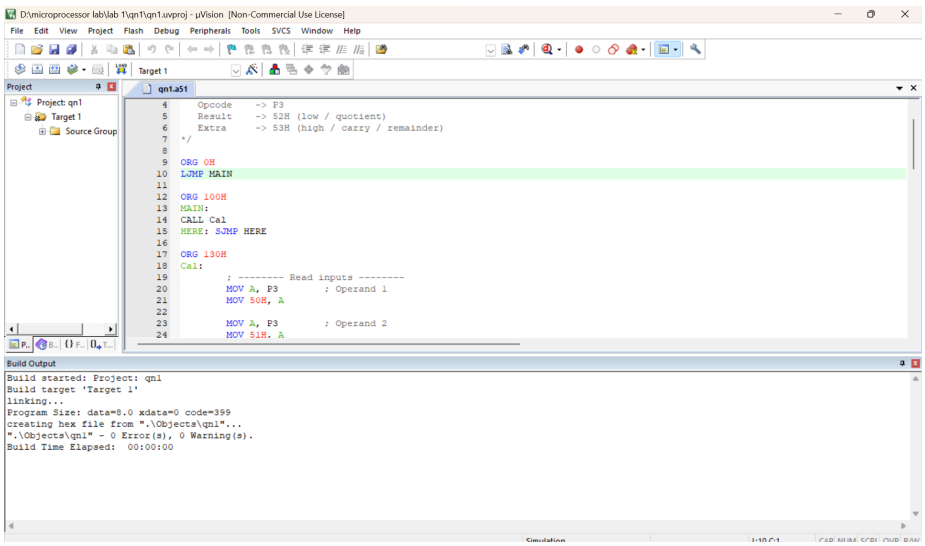
2 Problem 2

2.1 Objective

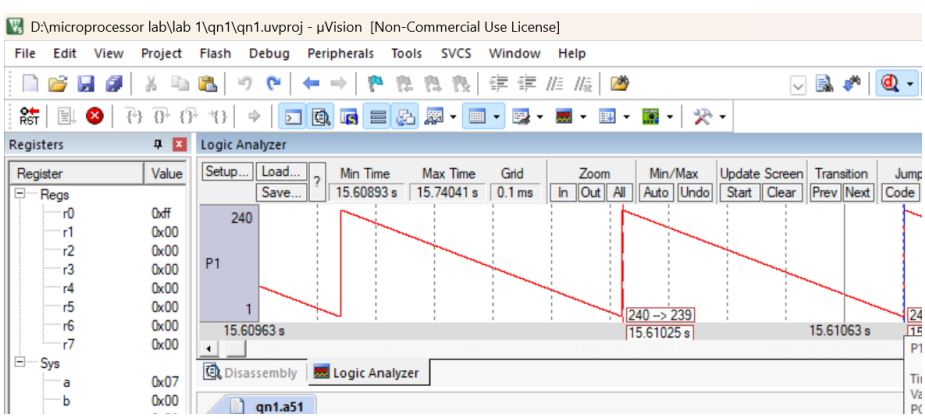
To debug the corrected 8051 assembly program using Keil µVision. To provide operand and opcode inputs through Port 3 using breakpoints.To observe how input data is stored in internal RAM.To analyze the output waveform generated on Port 1 using the Logic Analyzer and determine its frequency.

2.2 Simulations

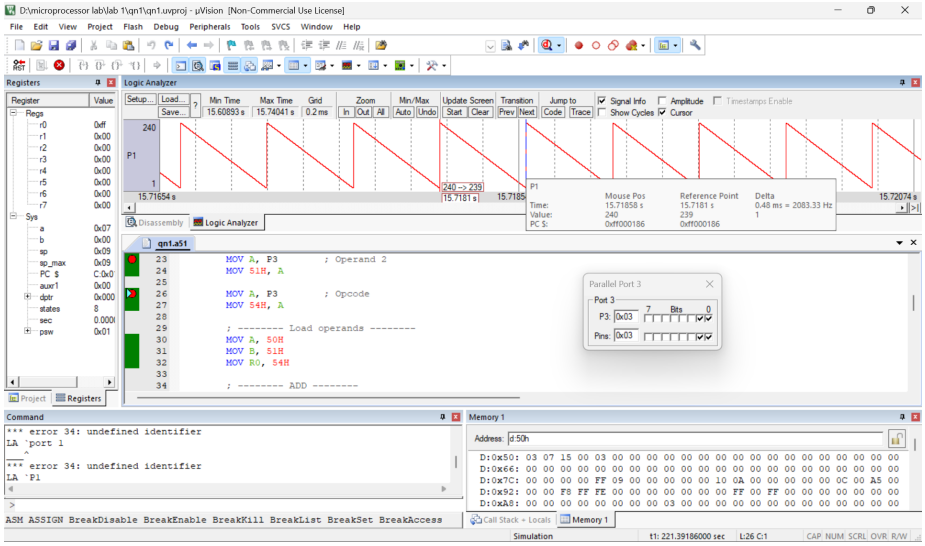
2.2.1 Build Successfully



2.2.2 Obtained Waveform



2.2.3 Debug Successfully with Obtained Frequency



2.3 Observations

- Breakpoints were successfully placed at Operand 1, Operand 2 and Opcode values were entered sequentially through the Port 3 peripheral window.
- The input values were correctly stored in internal memory locations 50H, 51H and 54H respectively and can be confirmed from memory window.
- After reaching the ENDING label, Port 1 continuously output a descending staircase (digital sawtooth) waveform.
- The Logic Analyzer showed a periodic waveform on Port 1.

2.4 Conclusion

- The observed waveform was a periodic descending staircase waveform, and its frequency was determined to be 2.1 kHz
- Based on the input to the opcode from 1 to 4 it performs operations Addition, Subtraction, Multiplication and Division.
- Here input to memory location 50H is 3, 51H is 7, opcode instruction is 3 that is Multiplication, therefore result 21 in Hexa 15 is obtained from 52H location.
- This experiment enhanced understanding of debugging techniques, I/O port operation, memory usage and waveform analysis in 8051 microcontroller programming.